

DIGITAL FORENSIC TRIAGE REPORT

Autonomous Incident Response & AI Threat Prioritization | ADCET Ashta

Institution:	Sant Dnyaneshwar Shikshan Sanstha's Annasaheb Dange College of Engineering and Technology (ADCET), Ashta
Department:	Department of Computer Science and Engineering (IoT & Cyber Security including Blockchain)
Project Guide:	Dr. Shabanam K. Shikalgar
Project Team:	Pragati Kiran Patil (23101024), Tanisha Prakash Lohar (23101009), Gaurav Subhash Rajguru (23101004)
Report Generated:	2026-08-20 21:27:45 UTC+05:30

1. Evidence Identification & Hash Verification

Target Filename	ransomware_incident_syslog.log	File Size	2,646 bytes
SHA-256 Hash	dd0be9cf2a87b54bed5cfee78b9f9dede56aa0b2419c69116f69fb1052a2f57d	File Type	LOG
MD5 Hash	615a5d1766f6695f2b4947584cf2a4f0	Acquisition Date	2026-08-18 00:02:14

2. Executive Summary & AI Risk Assessment

Overall Risk Score	100/100 (CRITICAL)	Primary Threat Vector	Ransomware / Extortion (86.7% conf)
AI Anomaly Status	Anomalous (Score: 56.01)	Detected Indicators	6 IOCs 12 Suspicious Findings

Investigative Summary: Automated digital forensic triage of 'ransomware_incident_syslog.log' identified a CRITICAL risk rating (Score: 100/100). The primary threat vector is classified as 'Ransomware / Extortion' with 86.7% confidence. Discovered 6 IOCs, 3 signature matches, and 5 associated MITRE ATT&CK techniques. AI Anomaly Engine marked system status as 'Anomalous'.

3. MITRE ATT&CK; Framework Correlation

Tactic	ID	Technique Name	Trigger / Evidence
Impact	T1490	Inhibit System Recovery	vssadmin
Impact	T1486	Data Encrypted for Impact	lockbit
Execution	T1059.001	Command and Scripting Interpreter: PowerShell	powershell
Initial Access	T1190	Exploit Public-Facing Application	rce
Defense Evasion	T1070.001	Indicator Removal: Clear Windows Event Logs	wevtutil

4. Key Indicators of Compromise (IOCs)

Type	Extracted Indicator (Defanged)	Classification / Context
IP Address	185[.]220[.]101[.]5	Public / External
IP Address	194[.]26[.]29[.]112	Public / External
URL	hxxp://185[.]220[.]101[.]5/payload/stage2[.]e	HTTP
SHA-256	e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855	Extracted Hash
MD5	7b2c9a6234b071e62a8291a13426e95c	Extracted Hash

LOLBAS Command	powershell.exe -nop -w hidden -enc	High
LOLBAS Command	vssadmin.exe delete shadows	Critical
LOLBAS Command	wmic shadowcopy delete	High
LOLBAS Command	wbadmin.exe delete catalog	High

5. Chronological Incident Timeline

Timestamp	Category	Severity	Event Snippet
2026-08-17 02:14:22	System Event	Info	Aug 17 02:14:22 DC01-SRV Microsoft-Windows-Securit
2026-08-17 02:14:45	Suspicious Execution / LOLBAS	High	Aug 17 02:14:45 DC01-SRV Microsoft-Windows-PowerSh
2026-08-17 02:15:10	Ransomware / Critical Threat	Critical	Aug 17 02:15:10 DC01-SRV Microsoft-Windows-Sysmon[
2026-08-17 02:15:33	Ransomware / Critical Threat	Critical	Aug 17 02:15:33 DC01-SRV Microsoft-Windows-Sysmon[
2026-08-17 02:16:01	Ransomware / Critical Threat	Critical	Aug 17 02:16:01 DC01-SRV Microsoft-Windows-Sysmon[
2026-08-17 02:16:08	System Event	Info	Aug 17 02:16:08 DC01-SRV Microsoft-Windows-Sysmon[
2026-08-17 02:16:15	System Event	Info	Aug 17 02:16:15 DC01-SRV Microsoft-Windows-Sysmon[
2026-08-17 02:16:22	System Event	Info	Aug 17 02:16:22 DC01-SRV Microsoft-Windows-Sysmon[

6. AI Containment Actions & Mitigation Playbook

- **[Network Containment] (High):** Block 2 malicious external IP(s) on perimeter firewall and EDR.
- **[Endpoint Isolation] (Immediate):** Immediately disconnect host from corporate LAN and Wi-Fi to prevent lateral encryption spreading.
- **[Backup Verification] (Immediate):** Check offline immutable backups and verify integrity of VSS shadow copies on unaffected systems.
- **[Persistence Removal] (Medium):** Audit and remove unauthorized Windows Registry Run keys, Scheduled Tasks, and anomalous startup services.

Forensic Checklist:

[] Inspect Master File Table (\$MFT) and USN Journal for mass file renaming activity.

[] Identify initial access vector (Phishing email, RDP brute force, or unpatched VPN/firewall).

[] Inspect AutoRuns / WMI event subscriptions for disguised binaries.

[] Preserve full volatile memory dump (.raw/.dmp) using WinPmem or FTK Imager prior to power down.

7. Chain of Custody & Digital Sign-off

I hereby certify that the digital evidence described above was processed under strict forensic protocols, and all extracted findings and cryptographic hashes were verified for integrity.

Lead Investigator:	_____	Signature:	_____
Project Guide:	Dr. Shabanam K. Shikalgar	Date:	20-08-2026
Verification SHA-256:	dd0be9cf2a87b54bed5cfee78b9f9dede56aa0b2...	Integrity Status:	VERIFIED / UNALTERED